

on every compact subset of Ω to a holomorphic function f , then f is either injective or constant.

Proof. We argue by contradiction and suppose that f is not injective, so there exist distinct complex numbers z_1 and z_2 in Ω such that $f(z_1) = f(z_2)$. Define a new sequence by $g_n(z) = f_n(z) - f_n(z_1)$, so that g_n has no other zero besides z_1 , and the sequence $\{g_n\}$ converges uniformly on compact subsets of Ω to $g(z) = f(z) - f(z_1)$. If g is not identically zero, then z_2 is an isolated zero for g (because Ω is connected); therefore

$$1 = \frac{1}{2\pi i} \int_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta,$$

where γ is a small circle centered at z_2 chosen so that g does not vanish on γ or at any point of its interior besides z_2 . Therefore, $1/g_n$ converges uniformly to $1/g$ on γ , and since $g'_n \rightarrow g'$ uniformly on γ we have

$$\frac{1}{2\pi i} \int_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta \rightarrow \frac{1}{2\pi i} \int_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta.$$

But this is a contradiction since g_n has no zeros inside γ , and hence

$$\frac{1}{2\pi i} \int_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta = 0 \quad \text{for all } n.$$

3.3 Proof of the Riemann mapping theorem

Once we have established the technical results above, the rest of the proof of the Riemann mapping theorem is very elegant. It consists of three steps, which we isolate.

Step 1. Suppose that Ω is a simply connected proper open subset of $\mathbb{C}$. We claim that Ω is conformally equivalent to an open subset of the unit disc that contains the origin. Indeed, choose a complex number α that does not belong to Ω , (recall that Ω is proper), and observe that $z - \alpha$ never vanishes on the simply connected set Ω . Therefore, we can define a holomorphic function

$$f(z) = \log(z - \alpha)$$

with the desired properties of the logarithm. As a consequence one has, $e^{f(z)} = z - \alpha$, which proves in particular that f is injective. Pick a point $w \in \Omega$, and observe that

$$f(z) \neq f(w) + 2\pi i \quad \text{for all } z \in \Omega$$

for otherwise, we exponentiate this relation to find that $z = w$, hence $f(z) = f(w)$, a contradiction. In fact, we claim that $f(z)$ stays strictly away from $f(w) + 2\pi i$, in the sense that there exists a disc centered at $f(w) + 2\pi i$ that contains no points of the image $f(\Omega)$. Otherwise, there exists a sequence $\{z_n\}$ in Ω such that $f(z_n) \rightarrow f(w) + 2\pi i$. We exponentiate this relation, and, since the exponential function is continuous, we must have $z_n \rightarrow w$. But this implies $f(z_n) \rightarrow f(w)$, which is a contradiction. Finally, consider the map

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}.$$

Since f is injective, so is F , hence $F : \Omega \rightarrow F(\Omega)$ is a conformal map. Moreover, by our analysis, $F(\Omega)$ is bounded. We may therefore translate and rescale the function F in order to obtain a conformal map from Ω to an open subset of $\mathbb{D}$ that contains the origin.

Step 2. By the first step, we may assume that Ω is an open subset of $\mathbb{D}$ with $0 \in \Omega$. Consider the family $\mathcal{F}$ of all injective holomorphic functions on Ω that map into the unit disc and fix the origin:

$$\mathcal{F} = \{f : \Omega \rightarrow \mathbb{D} \text{ holomorphic, injective and } f(0) = 0\}.$$

First, note that $\mathcal{F}$ is non-empty since it contains the identity. Also, this family is uniformly bounded by construction, since all functions are required to map into the unit disc.

Now, we turn to the question of finding a function $f \in \mathcal{F}$ that maximizes $|f'(0)|$. First, observe that the quantities $|f'(0)|$ are uniformly bounded as f ranges in $\mathcal{F}$. This follows from the Cauchy inequality (Corollary 4.3 in Chapter 2) for f' applied to a small disc centered at the origin.

Next, we let

$$s = \sup_{f \in \mathcal{F}} |f'(0)|,$$

and we choose a sequence $\{f_n\} \subset \mathcal{F}$ such that $|f'_n(0)| \rightarrow s$ as $n \rightarrow \infty$. By Montel's theorem (Theorem 3.3), this sequence has a subsequence that converges uniformly on compact sets to a holomorphic function f on Ω . Since $s \geq 1$ (because $z \mapsto z$ belongs to $\mathcal{F}$), f is non-constant, hence injective, by Proposition 3.5. Also, by continuity we have $|f(z)| \leq 1$ for all $z \in \Omega$ and from the maximum modulus principle we see that $|f(z)| < 1$. Since we clearly have $f(0) = 0$, we conclude that $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. In this last step, we demonstrate that f is a conformal map from Ω to $\mathbb{D}$. Since f is already injective, it suffices to prove that f is also surjective. If this were not true, we could construct a function in $\mathcal{F}$ with derivative at 0 greater than s . Indeed, suppose there exists $\alpha \in \mathbb{D}$ such that $f(z) \neq \alpha$, and consider the automorphism ψ_α of the disc that interchanges 0 and α , namely

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

Since Ω is simply connected, so is $U = (\psi_\alpha \circ f)(\Omega)$, and moreover, U does not contain the origin. It is therefore possible to define a square root function on U by

$$g(w) = e^{\frac{1}{2} \log w}.$$

Next, consider the function

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

We claim that $F \in \mathcal{F}$. Clearly F is holomorphic and it maps 0 to 0. Also F maps into the unit disc since this is true of each of the functions in the composition. Finally, F is injective. This is clearly true for the automorphisms ψ_α and $\psi_{g(\alpha)}$; it is also true for the square root g and the function f , since the latter is injective by assumption. If h denotes the square function $h(w) = w^2$, then we must have

$$f = \psi_\alpha^{-1} \circ h \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

But Φ maps $\mathbb{D}$ into $\mathbb{D}$ with $\Phi(0) = 0$, and is not injective because F is and h is not. By the last part of the Schwarz lemma, we conclude that $|\Phi'(0)| < 1$. The proof is complete once we observe that

$$f'(0) = \Phi'(0)F'(0),$$

and thus

$$|f'(0)| < |F'(0)|,$$

contradicting the maximality of $|f'(0)|$ in $\mathcal{F}$.

Finally, we multiply f by a complex number of absolute value 1 so that $f'(0) > 0$, which ends the proof.

For a variant of this proof, see Problem 7.

Remark. It is worthwhile to point out that the only places where the hypothesis of simple-connectivity entered in the proof were in the uses of the logarithm and the square root. Thus it would have sufficed to have assumed (in addition to the hypothesis that Ω is proper) that Ω is **holomorphically simply connected** in the sense that for any holomorphic function f in Ω and any closed curve γ in Ω , we have $\int_{\gamma} f(z) dz = 0$. Further discussion of this point, and various equivalent properties of simple-connectivity, are given in Appendix B.

4 Conformal mappings onto polygons

The Riemann mapping theorem guarantees the existence of a conformal map from any proper, simply connected open set to the disc, or equivalently to the upper half-plane, but this theorem gives little insight as to the exact form of this map. In Section 1 we gave various explicit formulas in the case of regions that have symmetries, but it is of course unreasonable to ask for an explicit formula in the general case. There is, however, another class of open sets for which there are nice formulas, namely the polygons. Our aim in this last section is to give a proof of the Schwarz-Christoffel formula, which describes the nature of conformal maps from the disc (or upper half-plane) to polygons.

4.1 Some examples

We begin by studying some motivating examples. The first two correspond to easy (but infinite and degenerate) cases.

EXAMPLE 1. First, we investigate the conformal map from the upper half-plane to the sector $\{z : 0 < \arg z < \alpha\pi\}$, with $0 < \alpha < 2$, given in Section 1 by $f(z) = z^{\alpha}$. Anticipating the Schwarz-Christoffel formula below, we write

$$z^{\alpha} = f(z) = \int_0^z f'(\zeta) d\zeta = \alpha \int_0^z \zeta^{-\beta} d\zeta$$

with $\alpha + \beta = 1$, and where the integral is taken along any path in the upper half-plane. In fact, by continuity and Cauchy's theorem, we may take the path of integration to lie in the closure of the upper half-plane. Although the behavior of f follows immediately from the original definition, we study it in terms of the integral expression above, since this provides insight for the general case treated later.

Note first that $\zeta^{-\beta}$ is integrable near 0 since $\beta < 1$, therefore $f(0) = 0$. Observe that when z is real and positive ($z = x$), then $f'(x) = \alpha x^{\alpha-1}$ is